

KIAN HEYDARI

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SUMMARY

Senior Developer specializing in **machine learning** and **generative AI applications**, backed by solid software and data engineering: **ETL pipelines**, **Azure** migration, and **CI/CD** in production, plus **agentic LLM pipelines** and **small language model distillation** in research. Works across **Python**, **SQL Server**, **Azure**, **Microsoft Fabric**, **Databricks**, **LangGraph**, **DSPy**, and **Hugging Face**.

EXPERIENCE

Senior IT Developer

May 2025 - Present

TD Securities, Credit Risk Reporting

Toronto, Canada

- Design, maintain, and optimize **ETL pipelines** in **Python** and **Microsoft SQL Server** that feed credit risk reporting.
- Migrated legacy ETL applications to **Azure**, replacing file-watcher triggers with **semantic dependencies between jobs**: an easier-to-follow flow, fewer failures, and **~50% fewer jobs**.
- Automate build, test, and deployment with **GitHub Actions CI/CD**, and keep code secure and maintainable with **Snyk**, **Veracode**, and **SonarQube**.
- Volunteer on the **ML Tooling** team, focusing on **agentic AI** and helping colleagues adopt the AI tools being rolled out. My team leads the organization by a **wide margin** on AI-integration dashboards.

Research Assistant (Part-time)

January 2025 - August 2025

Carleton University, supervised by Prof. Ali Arya

Ottawa, Canada

- Researched LLM translation for **low-resource Indigenous languages** such as Northern Tutchone, which mainstream models have seen little or no data for.
- Built a training-free **agentic translation pipeline** and a **knowledge distillation** pipeline for small, locally run translation models (see Projects).
- Wrote four literature reviews: LLM capabilities and efficiency; adapting generative models to low-resource languages; retrieving in-context examples without vector indexes; and grammar induction.
- Compared **mT5**, **mBART**, **XLM-R** adaptations (XLM-SWCM, XLM-V), **NLLB**, **BLOOMZ**, **ByT5**, language-specific (PhoBERT, AfriBERTa), and specialized models (FuxiMT, DeltaLM).

Analytics Engineer

February 2025 - April 2025

Revolution Data Platforms

Ottawa, Canada

- Developed a real-time customer service call management system using **Python**, **FastAPI**, and **WebSockets**, with end-to-end voice data pipelines.
- Prototyped voice memo transcription with **structured data extraction** and automated task management based on content severity.
- Optimized multi-class text classification through **prompt engineering** and **k-fold validation**.
- Improved data quality through **web scraping**, transformation, and preprocessing in Python.

Software Engineer

January 2023 - August 2024

Paphus Solutions

Ottawa, Canada

- Developed **chatbot APIs** across multiple platforms using Python and RESTful services.
- Optimized **deep learning** models for custom **question answering**.
- Integrated Python analytics pipelines with **cloud services** through optimized **ETL** workflows.
- Migrated a Unity-based Android application to WebGL with asynchronous connection management.

Course Instructor
Carleton University

September 2022 - December 2022
Ottawa, Canada

- Taught Python workshops on data manipulation, pandas, and ETL concepts.
- Guided students through ER modeling and SQL, emphasizing data integrity and query optimization.

Graduate Research Assistant
Carleton University

January 2021 - November 2022
Ottawa, Canada

- Built a Python data generation and transformation pipeline for deep learning research.
- Designed data preprocessing workflows and custom metrics for model evaluation.

Undergraduate Research Assistant
Amirkabir University of Technology

February 2018 - December 2020
Tehran, Iran

- Developed a Python anomaly detection system for time-series data, with feature extraction.
- Created simulation frameworks to generate and validate synthetic training data.

PROJECTS

- **Multi-Agent Translation System** *github.com/uni-VR-sity/translation-agent-system*
Training-free, few-shot translation into languages the base LLM was not trained on, inspired by “Hire a Linguist!”. Parallel agents pull relevant dictionary entries, grammar rules, and examples; a translator agent and an **LLM judge** run a self-correction loop. Adds a **map-reduce** mode for large documents and a **memory layer** that learns from judge feedback. Built with **LangGraph**, **DSPy**, **Instructor**, **LiteLLM**, and **Streamlit**.
- **Small Language Model Distillation for Translation** *PyTorch, Hugging Face, LoRA, MLflow*
Teacher LLMs (**Mistral**, **Gemma 2**) enrich a small English–Northern Tutchone parallel corpus with paraphrases, context, and longer sequences; multilingual students (**mT5**, **M2M100**, **mBART-50**) are fine-tuned fully or with **LoRA** in both directions, with BLEU-based early stopping, grid search, and MLflow tracking.
- **Fine-Tuning and Local RAG for LLMs**: lightweight RAG pipeline in Python for domain-specific data.
- **Custom Information Retrieval System**: search engine from scratch in Python, with custom indexing.
- **NLP-Driven Sentiment Analysis**: pandas/NumPy pipeline reaching 93.61% classification accuracy.

EDUCATION

M.Sc. in Computer Science (Specialization in Data Science)
Carleton University

January 2021 - November 2022
Ottawa, Canada

B.Sc. in Computer Engineering
Amirkabir University of Technology

September 2015 - December 2020
Tehran, Iran

SKILLS, LANGUAGES & CERTIFICATIONS

- **ML & Generative AI**: LangGraph, DSPy, Instructor, LiteLLM, Hugging Face Transformers, PyTorch, PEFT/LoRA, knowledge distillation, RAG, prompt engineering, MLflow, scikit-learn, pandas, NumPy.
- **Data Engineering**: ETL design and optimization, Microsoft SQL Server, Microsoft Fabric, Databricks, PostgreSQL, MySQL, NoSQL.
- **Cloud & DevOps**: Azure, AWS, GitHub Actions CI/CD, Docker, Git, Linux, Snyk, Veracode, SonarQube.
- **Backend & APIs**: FastAPI, Flask, Django, ASP.NET, REST, WebSockets, Streamlit.
- **Programming**: Python, SQL, C# (.NET), Java, JavaScript/TypeScript.
- **Languages**: English (Fluent), French (Intermediate), Persian (Native).
- **Certifications**: DataCamp Data Scientist with Python, Coursera GANs Specialization.